What Can Binary Response Patterns Reveal About Benchmark Contamination?

Characterizing the Minimal-Information Detection Frontier
with IRT Person-Fit Statistics

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Abstract

Benchmark contamination undermines the reliability of LLM evaluation, yet all existing detection methods require privileged access beyond binary (correct/incorrect) response patterns. We establish the *minimal-information operating point* on the contamination detection frontier by repurposing Item Response Theory (IRT) person-fit statistics—originally developed for exam-cheating detection—to identify aberrant LLM response patterns. Through controlled contamination experiments (QLoRA SFT injection at 0–50% levels) across three model families (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B-v0.3), we find that person-fit detection achieves |Cohen's d| $= 2.24$ on two of three families at 15% contamination with $n = 10$ seeds, but degrades to $|d| = 0.27$ on Mistral due to architecture-dependent memorization rates (17% vs. 38%). At item level, *all* binary-only methods—including oracle supervised classifiers—achieve near-random discrimination (AUROC ≤ 0.597), revealing that SFT contamination produces a diffuse signal detectable only through model-level aggregation. Post-RL (GRPO) training erases the contamination signal entirely ($|d| = 0.35$, $p = 0.38$ vs. clean control), delimiting the lower bound of what binary response patterns can reveal. These results constitute a systematic characterization of the detection frontier: we identify *where* binary-only detection succeeds (SFT, high-memorization architectures, large benchmarks), *where* it fails (low-memorization architectures, post-RL, small benchmarks), and *why* (memorization fraction, θ-shift confounds, IRT assumption violations).

1 Introduction

As large language models (LLMs) are increasingly evaluated on standardized benchmarks, the risk that training data overlaps with evaluation items—*benchmark contamination*—threatens the validity of reported performance gains. Detection methods have proliferated in response: n-gram overlap analysis requires access to training corpora; Min-K% and membership inference attacks (MIA) require token-level probabilities [Shi et al., 2024]; Time Travel probing requires crafted prompts [Dekoninck et al., 2024a]; ConStat requires controlled perturbations [Dekoninck et al., 2024b]. Critically, Wang et al. [2025] demonstrate that *all* MIA-based methods degrade to near-random performance on post-RL models, as reinforcement learning shifts token-level distributions.

A fundamental question remains unaddressed: **what can be detected from binary response patterns alone?** Binary patterns—correct/incorrect on each item—represent the minimal information available for any evaluated model, requiring no access to weights, logits, token probabilities, or training data. This minimal-information operating point defines the lower bound of the contamination detection frontier.

We answer this question by connecting to a mature psychometric tradition. In educational testing, *person-fit statistics* have been developed over decades to detect examinees with item preknowledge [Meijer & Sijtsma, 1996, Snijders, 2001]—precisely the analogue of benchmark contamination. The ℓ_z^* statistic [Snijders, 2001] measures whether an examinee's response pattern is aberrant relative to their estimated ability under an Item Response Theory (IRT) model: a contaminated model, like a cheating examinee, answers hard items correctly while missing easy ones, producing patterns inconsistent with its overall ability.

Our contributions are:

1. **Framework**: We formalize IRT person-fit as a contamination detection method operating at the minimal-information level, requiring only binary response patterns and pre-estimated item parameters. We introduce $\Delta\ell_z^*$ (within-model shift), θ-fixed analysis (controlling for ability confounds), and memorization fraction quantification.

2. **Detection Frontier Characterization**: Through controlled experiments across three model families, three benchmarks, six contamination levels, and two training paradigms (SFT, SFT+RL), we systematically map where binary-only detection succeeds and fails, identifying architecture-dependent memorization rate, benchmark size, and post-RL smoothing as the primary modulators.

3. **Methodological Insights**: We establish that (a) 4PL IRT parameterization yields $5.4\times$ the effect size of classical person-fit indices; (b) item-level detection is near-random for *all* binary-only methods, making model-level aggregation essential; (c) θ-fixed analysis recovers monotonic dose-response; and (d) post-RL training erases contamination signal at the response level, paralleling token-level findings of Wang et al. [2025].

2 Background

2.1 Item Response Theory

IRT models the probability that examinee j with ability θ_j answers item i correctly. The four-parameter logistic (4PL) model is:

$$P(X_{ij} = 1 \mid \theta_j) = c_i + (d_i - c_i) \cdot \frac{1}{1 + \exp(-a_i(\theta_j - b_i))} \tag{1}$$

where a_i is discrimination, b_i is difficulty, c_i is the guessing parameter, and d_i is the upper asymptote (slip parameter). We use 4PL parameters from PSN-IRT [PSN-IRT, 2026], which provides validated item parameters estimated from 1,034 models across multiple benchmarks.

2.2 Person-Fit Statistics

The ℓ_z^* statistic [Snijders, 2001] is the variance-corrected standardization of the log-likelihood of an observed response pattern:

$$\ell_z^* = \frac{\ell_0 - E[\ell_0]}{\sqrt{\text{Var}[\ell_0]}} \tag{2}$$

where $\ell_0 = \sum_i [x_{ij} \log P_i + (1 - x_{ij}) \log(1 - P_i)]$ is the log-likelihood, and expectation and variance are computed under the IRT model at $\hat{\theta}_j$. Under the null (response pattern consistent with the IRT model), $\ell_z^* \sim N(0, 1)$. Extreme negative values indicate *aberrant* patterns—unexpected successes on hard items and failures on easy items—the signature of contamination.

2.3 Contamination as Person Misfit

A contaminated model has been exposed to specific benchmark items during training. If this exposure produces *memorization* (item-specific recall rather than generalized capability), the resulting response pattern violates the monotonic ability-difficulty relationship assumed by IRT: the model answers memorized hard items correctly while still failing on non-memorized easy items. This aberrant pattern produces $\ell_z^* \ll 0$, analogous to an examinee with item preknowledge. We define the contamination detection signal as:

$$\Delta\ell_z^* = \ell_z^*(\text{contaminated}) - \ell_z^*(\text{clean baseline}) \tag{3}$$

3 Method

3.1 Controlled Contamination Protocol

We inject contamination via QLoRA supervised fine-tuning (SFT) on benchmark items at controlled levels $\gamma \in \{0\%, 5\%, 10\%, 15\%, 25\%, 50\%\}$. For each model family, we train n clean-seed models ($\gamma = 0\%$) to establish the null distribution of ℓ_z^* under QLoRA training noise alone, and compare against contaminated models. Effect size is measured as Cohen's d:

$$d = \frac{\ell_z^*(\text{contam}) - \overline{\ell_z^*}(\text{clean})}{\text{SD}_{\text{clean}}(\ell_z^*)} \tag{4}$$

3.2 θ-Fixed Analysis

Contamination can raise the maximum-likelihood ability estimate $\hat{\theta}$, partially absorbing the aberrance signal. We introduce θ-fixed analysis: recompute ℓ_z^* for contaminated models while fixing θ to the pre-contamination clean baseline $\bar{\theta}_{\text{clean}}$, isolating the pure item-pattern aberrance from ability shifts:

$$\ell_z^{*(\text{fixed})} = \ell_z^*(X_{\text{contam}} \mid \theta = \bar{\theta}_{\text{clean}}) \tag{5}$$

3.3 Memorization Fraction

We decompose contaminated items into *seen* (included in SFT training) and *unseen* subsets, defining the memorization fraction:

$$\text{MF} = \frac{\text{acc}_{\text{seen}} - \text{acc}_{\text{unseen}}}{\text{acc}_{\text{seen}}} \tag{6}$$

MF quantifies what proportion of accuracy gain on seen items reflects item-specific memorization rather than domain-level generalization.

3.4 Baselines

We compare against five baseline methods operating at or near the binary-information level: (1) HT person-fit [Harnisch, 1983]—an alternative IRT-based statistic without variance correction; (2) Guttman U3—a classical person-fit index using only item difficulty ordering; (3) accuracy z-score—standardized accuracy deviation; (4) PPL-based detection—perplexity-based method requiring token probabilities (above binary level); (5) random baseline. At item level, we additionally evaluate IRT residuals, logistic regression (LR) with cross-model features, and random forest (RF) with oracle contamination labels.

4 Experimental Setup

Models. Three 7B-parameter model families: Qwen2.5-7B-Instruct ($n = 10$ clean seeds), Llama-3.1-8B-Instruct ($n = 10$ clean seeds), and Mistral-7B-v0.3 ($n = 3$ clean seeds). All fine-tuned with QLoRA (rank 16, $\alpha = 32$, learning rate 2×10^{-4}).

Benchmarks. Primary: MMLU (14,042 items). Cross-benchmark: HellaSwag (10,042 items), GSM8K (1,319 items). Extended: BBH (6,511 items) and MATH (5,000 items) with self-calibrated 4PL parameters.

IRT Parameters. MMLU, HellaSwag, and GSM8K use PSN-IRT validated 4PL parameters [PSN-IRT, 2026]. BBH and MATH parameters are self-calibrated from Open LLM Leaderboard v1 data ($N \approx 120$ and $N \approx 80$ models, respectively).

Table 1: Detection power at 15% MMLU contamination. $|d|$ (n=3) shows pilot estimates; $|d|$ (n=10) shows expanded estimates with 95% bootstrap CIs. "Seeds $|d|<1$" counts individual seeds whose pairwise effect size falls below the large-effect threshold.

Family	n	**Clean $\bar{\ell}_z^*\pm$SD**	**ℓ_z^*@15%**	**$\lvert d\rvert$ (n=3)**	**$\lvert d\rvert$ (n=10)**	**95% CI**
Qwen2.5-7B	10	7.83 ± 1.78	3.85	3.42	2.24	[0.88, 3.60]
Llama-3.1-8B	10	-0.06 ± 1.31	-2.99	2.95	2.24	[0.84, 3.64]
Mistral-7B	3	5.73 ± 2.88	4.96	—	0.27	—

Table 2: Cross-family mechanism comparison at 15% MMLU contamination. MF = memorization fraction. Mistral's low MF and high clean variance jointly explain its non-detection.

Metric	**Qwen**	**Llama**	**Mistral**	**Interpretation**
Clean ℓ_z^* SD (n=3)	1.41	0.97	2.88	Mistral highest sensitivity to QLoRA init
MF@15%	0.38	0.25	0.17	Mistral memorizes least
Seen–unseen Δacc	0.15	0.11	0.051	Mistral aberrance weakest
θ-shift @15%	0.29	0.39	0.09	Mistral ability barely changes
$\lvert d\rvert$@15%	2.24	2.24	0.27	Mistral undetectable

Post-RL Experiment. On Qwen2.5-7B, we apply GRPO [GRPO, 2024] after SFT contamination (15%), with rule-based reward on held-out MMLU items, group size 8, learning rate 5×10^{-7}, KL coefficient 0.05, for 2,000 steps.

5 Results

5.1 Detection Across Model Families

Table 1 presents the core detection results with $n = 10$ clean seeds for Qwen and Llama, and $n = 3$ for Mistral. At 15% contamination on MMLU, both Qwen and Llama yield $|d| = 2.24$ (large effect), while Mistral shows $|d| = 0.27$ (negligible).

The expansion from $n = 3$ to $n = 10$ seeds reveals substantial QLoRA stochasticity: Qwen's clean ℓ_z^* standard deviation increases from 1.41 to 1.78, and Llama's from 0.97 to 1.31. This widens the null distribution, reducing $|d|$ by 34% for Qwen (3.42→2.24) and 24% for Llama (2.95→2.24). While both remain large effects, the 95% CI lower bounds (0.88 and 0.84) include medium-effect territory, and 3 of 10 Qwen seeds and 2 of 10 Llama seeds individually yield $|d| < 1$.

Mistral's failure traces to a fundamentally lower memorization rate. Table 2 compares the three families mechanistically.

5.2 Memorization Fraction Across Families

Table 3 shows memorization fractions (MF) across families and contamination levels. The cross-family spread of 21 percentage points at 15% contamination (Qwen 38% vs. Mistral 17%) indicates that memorization rate is architecture-dependent, not a fixed property of the contamination mechanism.

5.3 θ-Fixed Monotonicity Analysis

Table 4 presents θ-fixed analysis results. When ability is fixed to the clean baseline, both Qwen and Llama show monotonically increasing detection signal with contamination level (ratios of 3.12 and 2.87 between 50% and 15%), confirming that θ-shift absorbs part of the aberrance signal under free estimation. Mistral shows non-monotonic behavior even under θ-fixed, confirming that the weak signal is intrinsic rather than absorbed by ability re-estimation.

Table 3: Memorization fraction (MF) by family and contamination level.

Family	**MF@15%**	**MF@50%**	**Δ from mean**
Qwen	0.38	0.29	+11.3pp
Llama	0.25	0.22	−1.7pp
Mistral	0.17	0.14	−9.7pp
Cross-family spread	21pp	15pp	—

Table 4: θ-fixed analysis: detection signal ($\Delta\ell_z^{*(\text{fixed})}$) when ability is held at the clean baseline.

Family	θ_{fixed}	$\Delta\ell_z^*$**@15%**	$\Delta\ell_z^*$**@50%**	**Ratio**	**Monotonic?**
Qwen	−0.097	3.17	9.89	3.12	Yes
Llama	−0.666	0.114	0.327	2.87	Yes
Mistral	−0.44	−0.35	−0.61	1.74	No

5.4 Cross-Benchmark Detection Power

Table 5 reports $|d|$ at 15% contamination across benchmarks. Detection power is strongest on MMLU (14,042 items), moderate on HellaSwag (10,042 items), and weak on GSM8K (1,319 items). BBH and MATH, with self-calibrated parameters, show further degradation due to both fewer items and lower-quality parameter estimates (median discrimination $a = 0.15$ and $a = 0.11$, respectively, vs. $a = 0.45$ for PSN-IRT MMLU).

The power function relating benchmark size and item quality to detection power shows $R^2 = 0.65$ on PSN-IRT-validated benchmarks, degrading to $R^2 = 0.42$ when self-calibrated BBH and MATH are included (Table 6). This degradation is attributable to parameter estimation quality rather than method failure: self-calibration from $N \leq 120$ models for $> 5,000$ items yields systematically biased discrimination estimates.

5.5 Multiscale Cross-Benchmark Aggregation

We test whether cross-benchmark aggregation via the multiscale ℓ_{zm} statistic [Hong et al., 2021] improves detection when contamination targets a single benchmark. Table 7 shows that ℓ_{zm} *dilutes* the MMLU-specific signal by averaging with clean benchmarks, reducing $|d|$ by 19–31%. This expected result confirms that cross-benchmark aggregation is counterproductive for benchmark-specific contamination.

5.6 Baseline Comparison

Model-level detection. Table 8 compares model-level detection across methods. ℓ_z^* and HT person-fit achieve comparable performance (mean $|d|$ of 1.58 and 1.60, respectively), both substantially outperforming Guttman/U3 ($|d| = 0.29$), confirming that full IRT parameterization provides a $5.4\times$ advantage over difficulty-only classical indices. PPL-based detection, which uses token probabilities (above the binary-information level), achieves only $|d| = 0.50$. Accuracy z-score shows extreme sensitivity (up to $z = 92.4$) but lacks specificity, as any fine-tuning—contaminated or not—inflates accuracy.

The near-equivalence of ℓ_z^* and HT ($\Delta|d| = 0.02$) indicates that the contribution lies in the *IRT person-fit family* rather than any specific statistic. ℓ_z^*'s theoretical advantage is the variance correction of Snijders [2001], which provides better-calibrated null distributions; in practice, both statistics leverage the same IRT-parameterized item information.

Item-level detection. Table 9 shows item-level AUROC for distinguishing contaminated from clean items. All methods achieve near-random discrimination (maximum AUROC = 0.597 for oracle random

Table 5: $|d|$ at 15% contamination across benchmarks (Qwen/Llama n=10, Mistral n=3). 4PL source indicates parameter estimation origin. a_{med} = median item discrimination.

Benchmark	n_{items}	**4PL Source**	a_{med}	**Qwen**	**Llama**	**Mistral**
MMLU	14,042	PSN-IRT	0.45	2.24	2.24	0.27
HellaSwag	10,042	PSN-IRT	0.38	1.35	1.62	0.31
BBH	6,511	Self-cal.	0.15	0.72	0.85	—
MATH	5,000	Self-cal.	0.11	0.48	0.41	—
GSM8K	1,319	PSN-IRT	0.32	0.58	0.73	0.12

Table 6: Power function R^2 degradation as benchmarks with lower-quality parameter estimates are included.

Benchmark Set	n_{points}	R^2	**RMSE**
PSN-IRT validated only	5	0.65	0.075
+ BBH	6	0.54	0.112
+ BBH + MATH	7	0.42	0.148

forest on Llama), confirming that SFT contamination produces a *diffuse* signal: with memorization fractions of 17–38%, only a minority of injected items develop detectable aberrance at the individual level. This fundamentally motivates model-level aggregation, where IRT provides principled weighting through item information functions.

5.7 Post-RL Detection Failure

Table 10 presents the central negative result. After GRPO training (2,000 steps) on an SFT-contaminated Qwen model, the ℓ_z^* detection signal drops from $|d| = 2.24$ to $|d| = 0.35$, statistically indistinguishable from the clean+GRPO control ($|d| = 0.21$, $p = 0.38$). Memorization fraction collapses from 38% to 6%, and the seen–unseen accuracy gap shrinks from 15.0pp to 1.5pp.

The mechanism is threefold: (1) GRPO's KL penalty ($\beta = 0.05$) pulls the policy toward the pre-SFT reference, systematically decaying item-specific memories; (2) group-based optimization ($n = 8$ samples per prompt) smooths aberrant response patterns by rewarding consistent correctness across items; (3) the reward signal (held-out accuracy) provides no reinforcement for memorized items specifically, allowing memorization to decay while genuine capability is reinforced.

Table 11 contextualizes this with prior work. Both response-level (ℓ_z^*) and token-level (Min-K%, MIA-LiRA) detection methods fail post-RL, with our method retaining only 15.6% of the pre-RL signal compared to $\sim$3–5% for token-level methods. This convergence across information levels suggests that RL fundamentally restructures both token distributions and response patterns, rendering contamination traces undetectable by any distributional method.

6 Discussion

6.1 The Detection Frontier

Our results map the boundaries of what binary response patterns can reveal about benchmark contamination, identifying three primary modulators:

Architecture-dependent memorization. The 21pp spread in memorization fraction across families (Qwen 38%, Llama 25%, Mistral 17%) indicates that contamination detection power is jointly determined by the detection method *and* the target architecture's propensity for item-specific memorization

Table 7: Multiscale ℓ_{zm} vs. per-benchmark ℓ_z^* at 15% MMLU contamination. Aggregating across benchmarks dilutes the MMLU-specific signal.

Method	**Qwen** $\lvert d\rvert$	**Llama** $\lvert d\rvert$	**Mistral** $\lvert d\rvert$
MMLU ℓ_z^* only	2.24	2.24	0.27
ℓ_{zm} (MMLU+GSM8K+HellaSwag)	1.82	1.55	0.22
Δ (dilution)	-19%	-31%	-19%

Table 8: Model-level detection at 15% MMLU contamination. $|d|$ reported; acc z-score is a z-statistic (not Cohen's d). CI width is bootstrap 95% CI half-width for IRT methods.

Method	**Info Level**	**Qwen**	**Llama**	**Mistral**	**Mean**	**CI**
ℓ_z^* (ours)	Binary	2.24	2.24	0.27	1.58	$\pm$1.36
HT person-fit	Binary	2.31	2.08	0.41	1.60	$\pm$1.28
Guttman/U3	Binary	0.28	0.51	0.09	0.29	$\pm$0.31
Acc z-score	Binary	92.4	68.3	25.7	62.1	—
PPL-based	Token-prob	0.62	0.71	0.18	0.50	$\pm$0.22
Random	—	0.00	0.00	0.00	0.00	—

vs. generalized capability improvement. Mistral's sliding window attention appears to promote diffuse learning that distributes contamination signal across the response pattern, making aberrance undetectable.

Benchmark characteristics. Detection power correlates with benchmark size and item discrimination quality. The power function ($R^2 = 0.65$ on validated parameters) provides practitioners approximate guidance: benchmarks with $\geq$10K well-discriminating items (e.g., MMLU, HellaSwag) support detection, while smaller or poorly discriminating benchmarks (GSM8K, MATH) fall below practical thresholds. IRT assumption violations—particularly local independence violations in chain-of-thought benchmarks (MATH)—further degrade detection.

Training paradigm. Post-RL training constitutes an absolute boundary: RL erases contamination signal at both the token level [Wang et al., 2025] and the response level (this work). Any method relying on distributional evidence—whether token probabilities or binary patterns—will fail on post-RL models. Only methods with access to pre-RL checkpoints or structural model probing can potentially succeed in this regime.

6.2 Methodological Contributions

4PL parameterization. The $5.4\times$ effect-size advantage of IRT-based person-fit (ℓ_z^*, HT) over classical Guttman/U3 validates the use of full 4PL item parameters. The upper-asymptote parameter d_i is critical for high-ability models where slip rates vary across items.

Model-level aggregation. The near-random item-level AUROC (max 0.597 even with oracle supervision) establishes that contamination signal is fundamentally diffuse under SFT. With memorization fractions of 17–38%, only a minority of injected items develop item-specific aberrance; the remainder produce generalized capability improvement indistinguishable from genuine learning. IRT provides principled aggregation: each item contributes according to its Fisher information, enabling detection from thousands of individually uninformative signals.

θ-shift as confound. The θ-fixed analysis reveals that ability re-estimation on contaminated data absorbs part of the aberrance signal. Fixing θ to the clean baseline recovers monotonic dose-response in

Table 9: Item-level AUROC at 15% contamination. Even oracle-supervised methods (RF with ground-truth labels) barely exceed random, confirming contamination signal is diffuse.

Method	Qwen	Llama	Mistral	Mean
ℓ_z^* itemwise	0.528	0.521	0.506	0.518
IRT residual	0.568	0.572	0.515	0.552
LR cross-model	0.579	0.584	0.521	0.561
RF oracle	0.592	0.597	0.531	0.573
Acc z-score (item)	0.541	0.533	0.509	0.528
Random	0.500	0.500	0.500	0.500

Table 10: Post-RL detection results on Qwen2.5-7B. GRPO erases the contamination signal: the SFT+GRPO condition is indistinguishable from the clean+GRPO control. MF = memorization fraction.

Condition	ℓ_z^*	$\Delta\ell_z^*$	$\lvert d\rvert$	MF	$\Delta\text{acc}_{\text{s-u}}$
Clean (n=10 mean)	7.83	—	—	—	0pp
SFT-only 15%	3.85	-3.98	2.24	0.38	15.0pp
SFT+GRPO 15%	7.21	-0.62	0.35	0.06	1.5pp
Clean+GRPO	7.45	-0.38	0.21	—	0pp

2/3 families, with the non-recovering family (Mistral) exhibiting intrinsically weak signal rather than θ-absorption. This identifies a practical consideration: detection should use pre-contamination ability baselines when available.

6.3 Limitations

QLoRA-specific findings. All controlled experiments use QLoRA SFT for contamination injection. Full fine-tuning, different LoRA configurations, or continued pretraining may produce different memorization fractions and detection characteristics.

IRT parameter dependency. Detection power depends critically on 4PL parameter quality. PSN-IRT parameters yield strong results; self-calibrated parameters from small respondent pools ($N \leq 120$) degrade detection substantially (R^2 from 0.65 to 0.42). The power function should not be extrapolated to benchmarks without validated parameters.

Sample size. While $n = 10$ clean seeds provide more stable estimates than $n = 3$, the 95% CI half-width (± 1.36) remains large. Individual seeds can yield $|d| < 1.0$, indicating that single-seed detection is unreliable.

Scope. We target intentional SFT contamination (leaderboard gaming). Incidental pretraining exposure produces smooth distributional improvement without item-specific memorization, and falls outside the detection capability of any person-fit method.

7 Related Work

Contamination detection. Shi et al. [2024] introduce Min-K% for membership inference using token probabilities. Dekoninck et al. [2024a] propose Time Travel probing with rephrased items. Dekoninck et al. [2024b] present ConStat, testing consistency under performance-level constraints. Wang et al. [2025] demonstrate that all MIA-based methods fail on RL-trained models. All require information beyond binary response patterns.

Table 11: Post-RL signal erasure across detection paradigms. Asterisked values are approximate from Wang et al. [2025]. Both response-level and token-level methods fail, delimiting the detection frontier's lower bound.

Method	Level	Pre-RL $\|d\|$	Post-RL $\|d\|$	Retention	Mechanism
ℓ_z^* (ours)	Response	2.24	0.35	15.6%	Pattern smoothed
Min-K%	Token	$\sim$2.0*	$\sim$0.1*	$\sim$5%*	Distribution shifted
MIA-LiRA	Token	$\sim$1.5*	$\sim$0.05*	$\sim$3%*	Membership signal lost

IRT for LLMs. PSN-IRT [2026] provide the PSN-IRT framework, estimating 4PL parameters from large-scale LLM response data. Prior work applies IRT to LLM evaluation for ability estimation and benchmark design, but none addresses contamination detection through person-fit statistics.

Person-fit in psychometrics. Meijer & Sijtsma [1996] survey person-fit statistics for detecting aberrant response patterns. Snijders [2001] derives the variance-corrected ℓ_z^* statistic. Hong et al. [2021] extend to multiscale aggregation (ℓ_{zm}). We are the first to apply person-fit statistics to LLM contamination detection.

8 Conclusion

We have established the minimal-information operating point on the contamination detection frontier. Using only binary response patterns, IRT person-fit statistics detect SFT contamination with large effect sizes ($|d| = 2.24$) on two of three model families, but this detection is modulated by architecture-dependent memorization rates, benchmark size, and parameter quality. Post-RL training erases the contamination signal entirely, delimiting an absolute boundary for response-pattern-based detection.

Rather than claiming a universal detection tool, we contribute a systematic characterization: person-fit statistics succeed when contamination produces item-specific memorization in sufficiently large benchmarks with well-estimated IRT parameters, and fail otherwise. The failure modes—low memorization architectures, post-RL smoothing, small benchmarks, poor parameter estimates—are themselves informative, defining the axes along which the detection frontier varies.

The item-level finding that *no* binary-only method exceeds AUROC 0.60 even with oracle supervision, while model-level aggregation yields $|d| > 2$, establishes a fundamental property of SFT contamination: it produces diffuse statistical signatures that require aggregation over thousands of items for reliable detection.

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